Practical exercise 8

1. Consider data on claim numbers for 55 car insurance policies:

Policyholder	numclaims(y)	age	sex	cargroup
1	0	17	m	A
2	0	19	f	A
.	.	.	.	.
55	3	32	m	B

The full data are given on Canvas in file `carclaims.R`. Save the file on your computer and read the data into `R` as a dataframe using a command similar to the following:

```
carclaims.df <- read.table("carclaims.R",header=T)
```

We consider the response variable `numclaims` (number of claims in a year) to depend on explanatory variables: `age` – age of policyholder; `sex` – sex of policyholder (declared as a factor, male or female); and `cargroup` – a factor in three categories A,B,C.

We assume that the number of claims experienced by policyholder i has a Poisson(λ_i) distribution where the mean λ_i depends on the above rating factors as follows:

$$\eta_i \;=\; \log\lambda_i \;=\; \beta_0 + \beta_1 x + \alpha I\{\texttt{sex} = \text{male}\} + \gamma_2 I\{\texttt{cargroup} = \text{B}\} + \gamma_3 I\{\texttt{cargroup} = \text{C}\}$$

This can be interpreted as follows:

- Age is a numerical variable x with values $17, 18, 19, \ldots, 84$; we include it in the linear predictor with a term $\beta_1 x$. The predictor, including an intercept term, is written
  $\eta = \beta_0 + \beta_1 x$
- Sex is a factor with two levels: base level (level 0) for female and level 1 for male; we allow for this by including a term α and we now write the predictor as
  $\eta = \beta_0 + \beta_1 x + \alpha I\{\texttt{sex} = \text{male}\}$
  where α is the effect for `sex` = male.
- Car Group is a factor with 3 levels; we allow for this by regarding our base level (level 1) to be car group A, and then including parameters γ_2 and γ_3 for car groups B and C respectively. Our predictor now appears as
  $\eta = \beta_0 + \beta_1 x + \alpha I\{\texttt{sex} = \text{male}\} + \gamma_2 I\{\texttt{cargroup} = \text{B}\} + \gamma_3 I\{\texttt{cargroup} = \text{C}\}$
  where γ_2 is the effect for car group B and γ_3 the effect for car group C.

(a) Write down expressions for the predictor η_i (in terms of the equation above) for the following cases:

- a 36 year-old female with car in group A
- a 24 year-old male with car in group C
- a 53 year-old female with car in group B.

(b) Fit the following models in `R` (in the provided order) and verify the deviance changes as given below:

- Model 1: `age`. Deviance = 61.146 on 53 df. Reduction in deviance compared to null model = 5.16 on 1 df.
- Model 2: `sex`. Deviance= 66.301 on 53 df. Change in deviance compared to null model = 0.005 on 1 df.
- Model 3: `cargroup`. Deviance =56.168 on 52 df. Change in deviance compared to null model=10.138 on 2 df.
- Model 4: `cargroup + sex`. Deviance = 56.136 on 51 df. Reduction in deviance compared to model 3 = 0.032 on 1 df.
- Model 5: `cargroup + age`. Deviance = 50.208 on 51 df. Reduction in deviance compared to model 3 = 5.96 on 1 df.
- Model 6: `cargroup + age + sex`. Deviance = 50.205 on 50 df. Reduction in deviance compared to model 5 = 0.003 on 1 df.

Compute the p-value associated with the reduction in deviance in each step, e.g. for Model 1,

```
1 - pchisq(5.16, df = 1)
```

Interpret this p-value for the change in Deviance for each model in the sequence of nested models fitted before.

Explain why, on this basis alone and when exploring the models in this sequence, we choose Model 5 as the most appropriate model.

We can perform these types of comparisons in `R`. For example use the command below to confirm some of your previous computations:

```
anova(m3, m5, test="Chi")
```

(c) Using the output from `R`, examine the significance of the parameters in Model 5.

Check the fit of the model using a chi-squared test involving the deviance.

(d) Calculate (manually) the linear predictor for the 3 cases in part (a) under Model 5, and hence give the estimated expected number of claims for each case.

You can find these values using the `predict()` command in `R`. For example verify your calculations for the first case using:

```
predict(m5, data.frame(age = 36, sex="f", cargroup="A"), type="link")
predict(m5, data.frame(age = 36, sex="f", cargroup="A"), type="response")
```

(e) Produce plots of the Pearson residuals of the model against the fitted values and age.

Does the fit of the model appear to be good?

2. For the model of Q1 of practical exercise 7 (log-linear Poisson regression model), show (algebraically) that the deviance is given by

$$D = -2\sum_{i=1}^{18}\left(y_i \log\left(\hat{\lambda}_i/y_i\right) + y_i - \hat{\lambda}_i\right)$$

where $\hat{\lambda}_i$ are the fitted values.

Use this to compute manually the deviance, and verify its value using the output from `R` obtained in Q1.

3. Consider again the data on the IQ of twins in Practical sheet 6, Q4. We want to fit a normal GLM with identity link, i.e.

$$\begin{aligned} Y_i &\sim N(\mu_i, \sigma^2) \\ \mu_i &= \eta_i = \boldsymbol{x}_i^T \boldsymbol{\beta} \end{aligned}$$

with an appropriate linear predictor (covariates).

(a) Show algebraically that the deviance of this model is given as

$$D = \sum_{i=1}^{n} (y_i - \hat{\mu}_i)^2 \Big/ \phi \qquad \text{with } \phi = \sigma^2$$

where $\hat{\mu}_i$ are the model fitted values.

(b) Fit the same model as in part (c) of Q4, Practical 4 (`Foster` $\sim$ `Biological+Social`), but using the `glm()` command. Confirm that the two commands (`lm` and `glm`) give the same results.

Confirm the estimate of ϕ using the `R` output from both the `lm()` and `glm()` commands. Also compute the Deviance using the expression in part (a) and then use the deviance from the `glm` output to confirm your answer.